Master’s thesis defense · DTU Electro

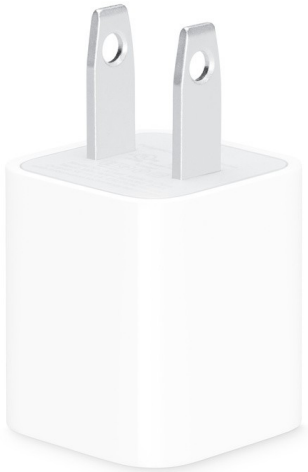
Design and Implementation of a High-Frequency Charge-Pump PFC Resonant Converter for a USB-C Power Supply



Richard Xu

More power from a compact USB-C adapter

Motivation



5 W USB adapter



20 W USB-C adapter

140 W USB-C adapter

Project target: 100 W continuous

Minimum: 65 W continuous

Examples shown at different scales · Apple

High frequency makes switching loss a design constraint

Motivation

Higher switching frequency → less energy per cycle

Smaller passives

Less energy per cycle can reduce magnetic and capacitor size

➡

More switching heat

$$P_{sw} \approx f_s E_{sw}$$

➡

Zero-voltage switching

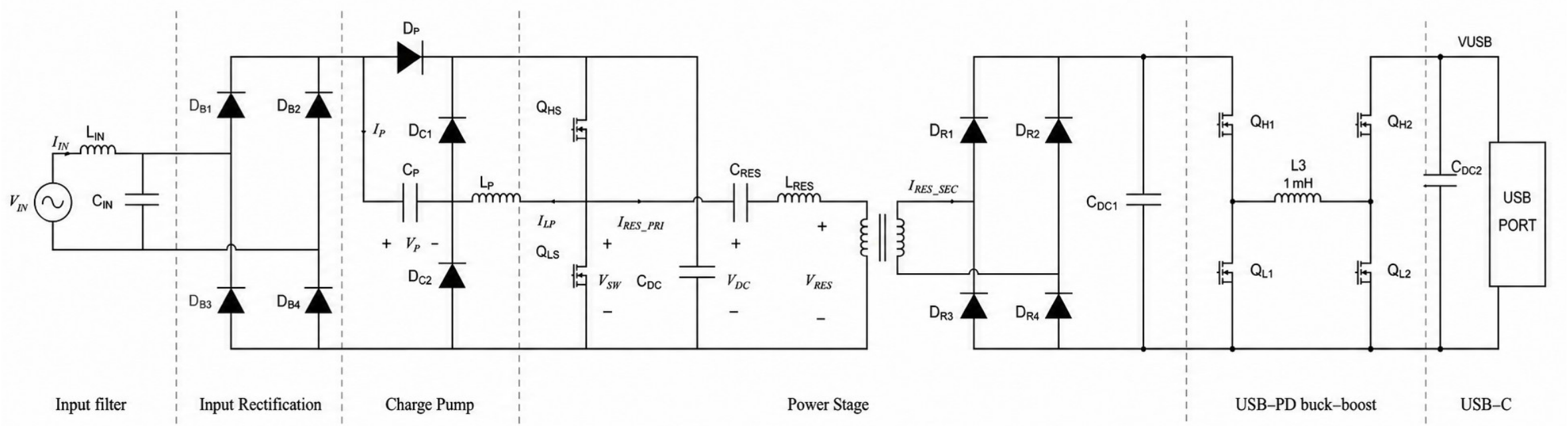
Resonant current lowers device voltage before turn-on

Changing USB-C demand still requires energy regulation

One half bridge drives both power paths

Circuit

Power stage



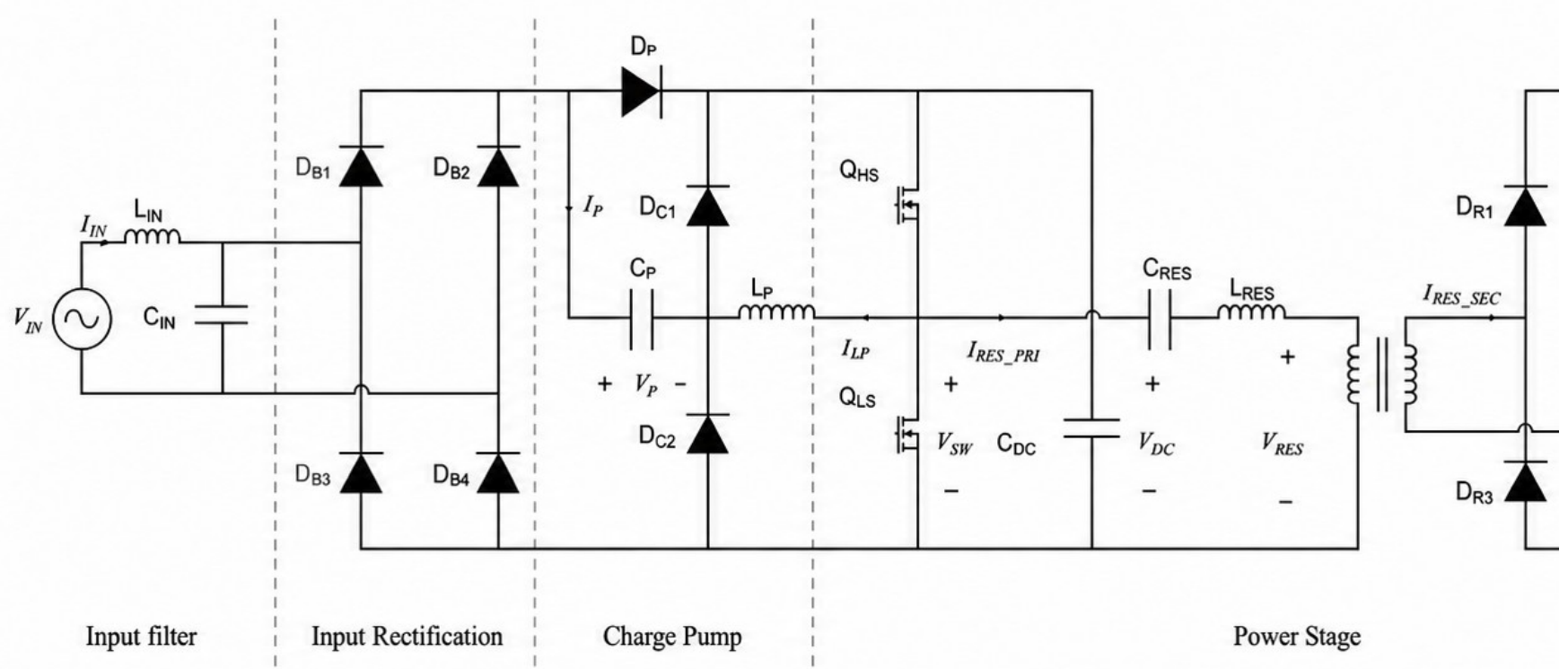
Project contribution

Sensing, local supplies and USB-C integration

Secondary-voltage burst control and prototype tests

Charge transfer shapes the input current

Circuit



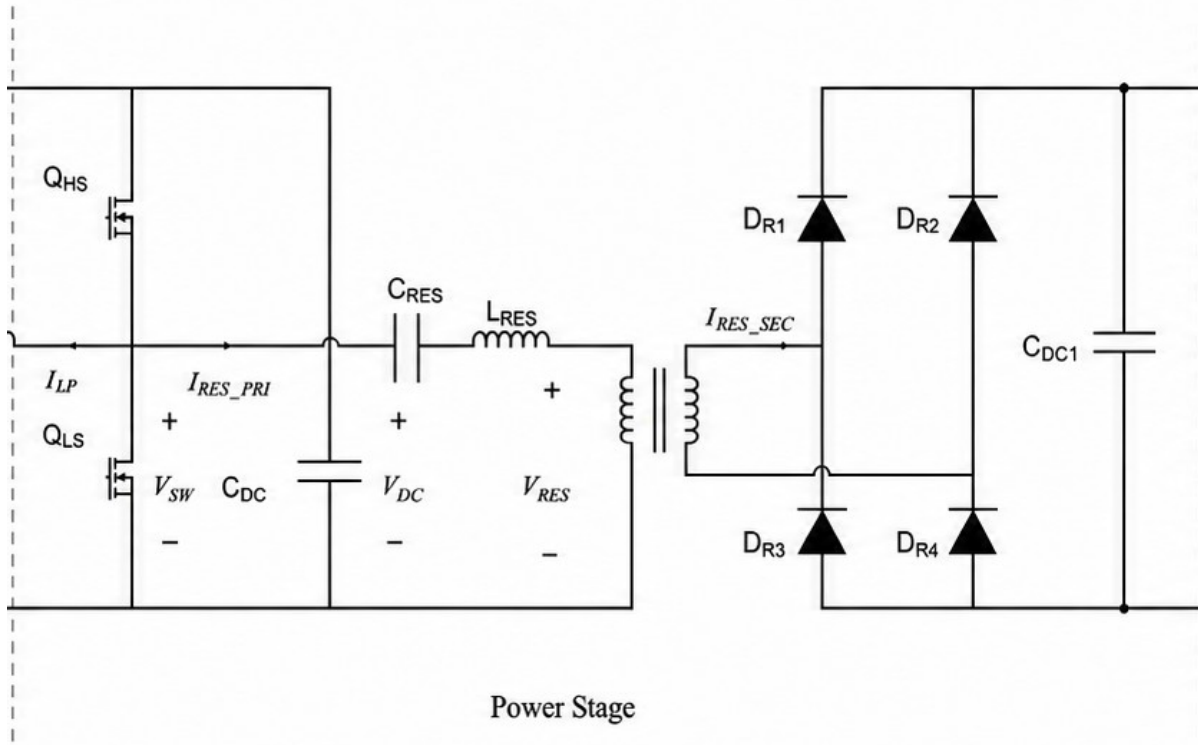
$$\bar{i}_{\mathrm{rec}} \approx b\, f_{\mathrm{s}}\, C_{\mathrm{P}}\, v_{\mathrm{rec}}$$

b: local fraction of enabled switching cycles

Inherent current shaping without a separate PFC current loop

Resonant transfer and shared-node commutation

Circuit



Near 1 MHz: isolated power transfer

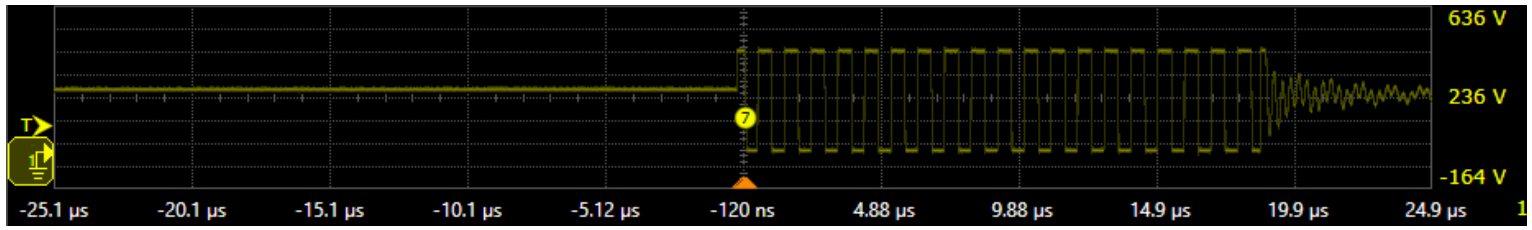
Pump + resonant currents move V_{SW} during dead time

Burst control regulates the secondary bulk

Control



V1 · 5 V / 3 A USB-C output



Idle interval

Switching packet

5 µs/div · 100 V/div

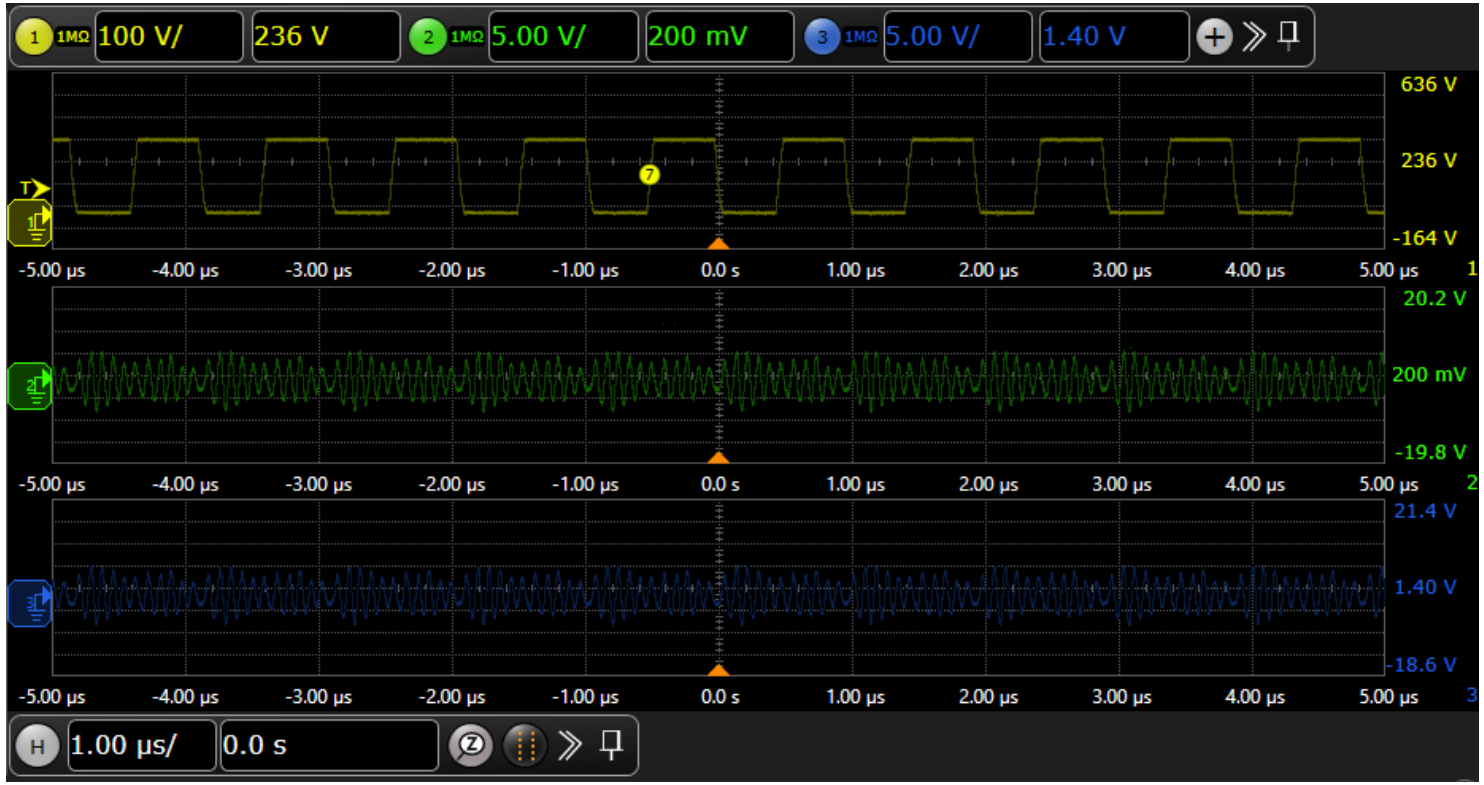
Yellow: half-bridge switch-node voltage

- Bulk voltage sets burst demand
- Nominal 1 MHz within each packet
- Startup and low-load energy is bounded

V1 switching near 1 MHz

Results

V1 · 20 V USB-C setting · 4.7 A load setting



About 10 cycles in 10 µs → approximately 1 MHz

Yellow: switch node · DC-coupled · 100 V/div

Green: secondary bulk · AC-coupled · 5 V/div

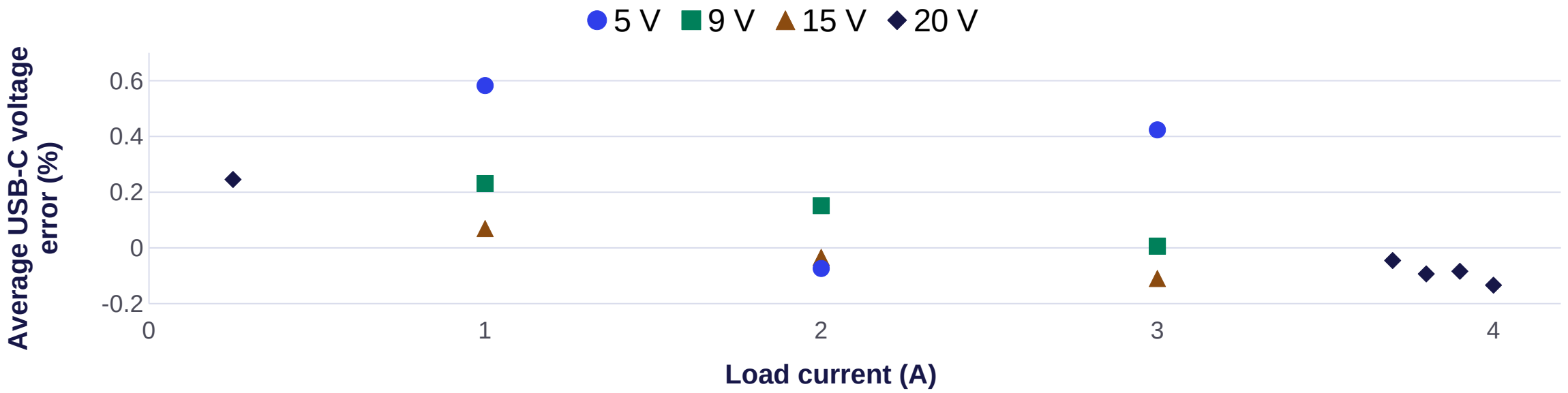
Blue: USB-C VBUS · AC-coupled · 5 V/div

Time: 1 µs/div

USB-C output stays close to the selected voltage

Results

V1 · 240 V AC, 50 Hz · USB-C connector



Four fixed settings: 5, 9, 15 and 20 V

Largest absolute average error: 0.583% at 5 V, 1 A

79.89 W at the USB-C connector

Efficiency and heat at 79.89 W

Results

V1 · 240 V AC, 50 Hz · 20 V selected

Input: 104.42 W → USB-C connector: 79.89 W

19.9732 V × 4.0 A

Efficiency estimate: 76.51%

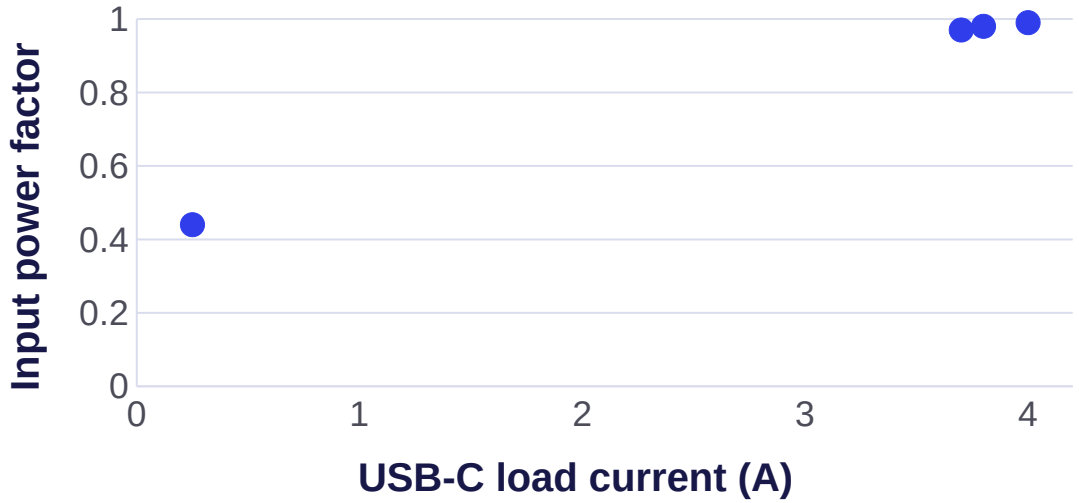
Input–output difference: 24.53 W

Good input quality at high load

Results

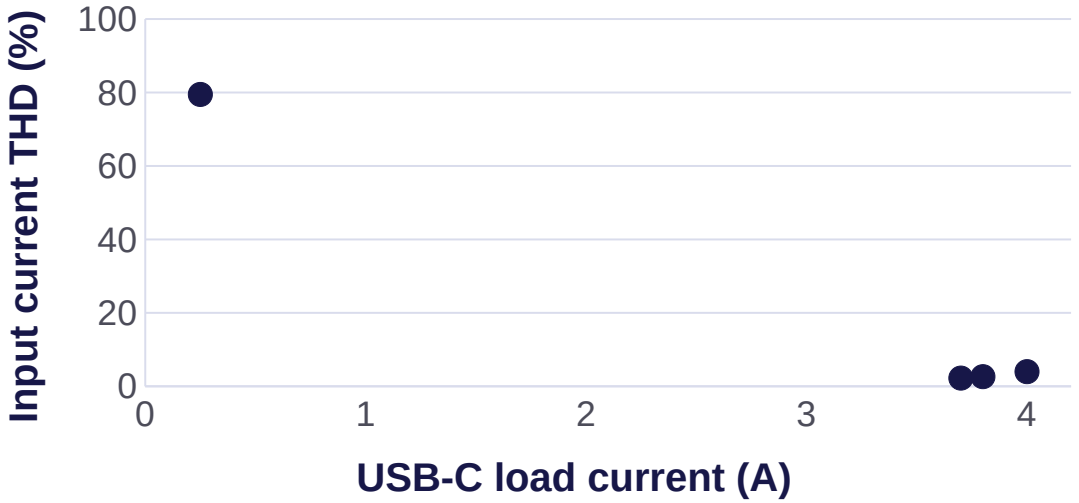
V1 · Selected USB-C output: 20 V

Input power factor



0.44 → 0.99

Input current THD (%)



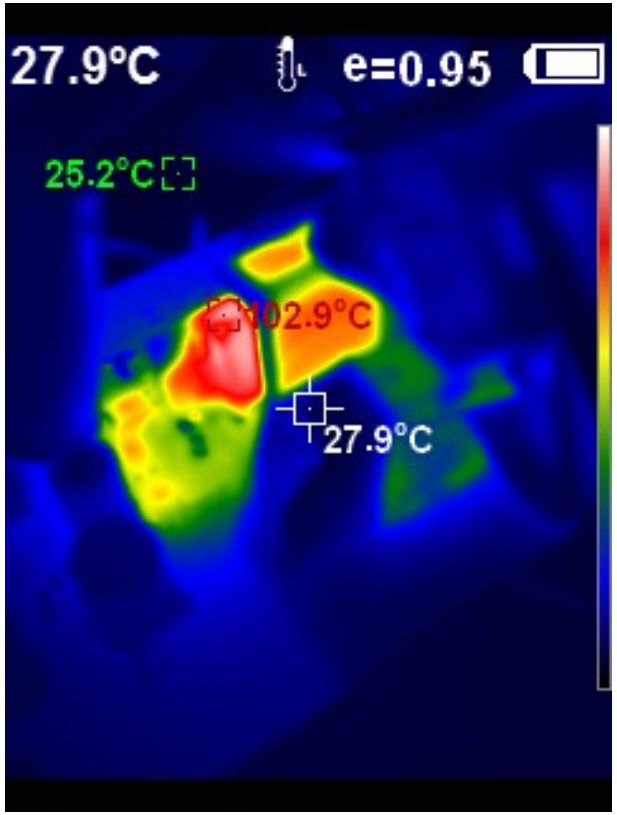
79.46% → 3.98%

Burst density can change the input-current envelope

V1 magnetics heating at high load

Results

V1 · 20 V, 4.7 A load setting



102.9 °C

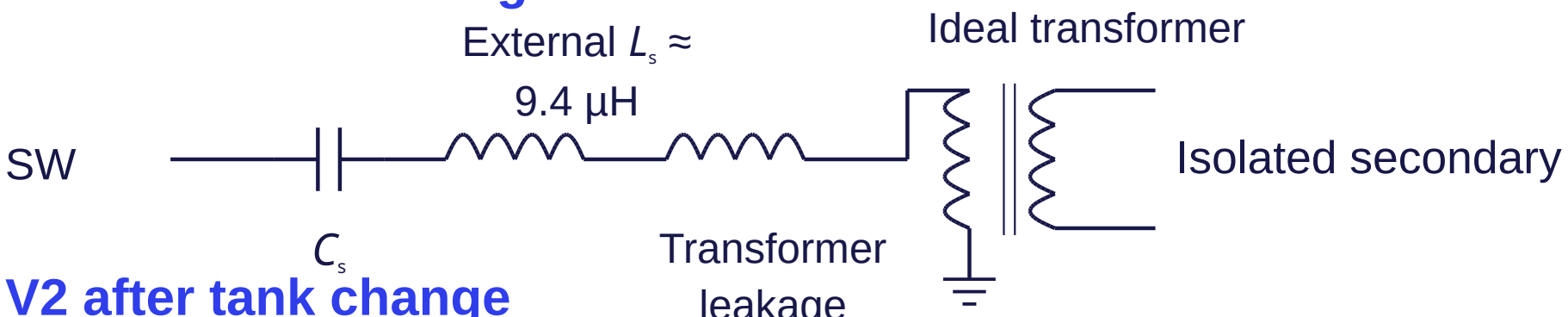
Maximum surface temperature

Heat concentrated around the magnetics

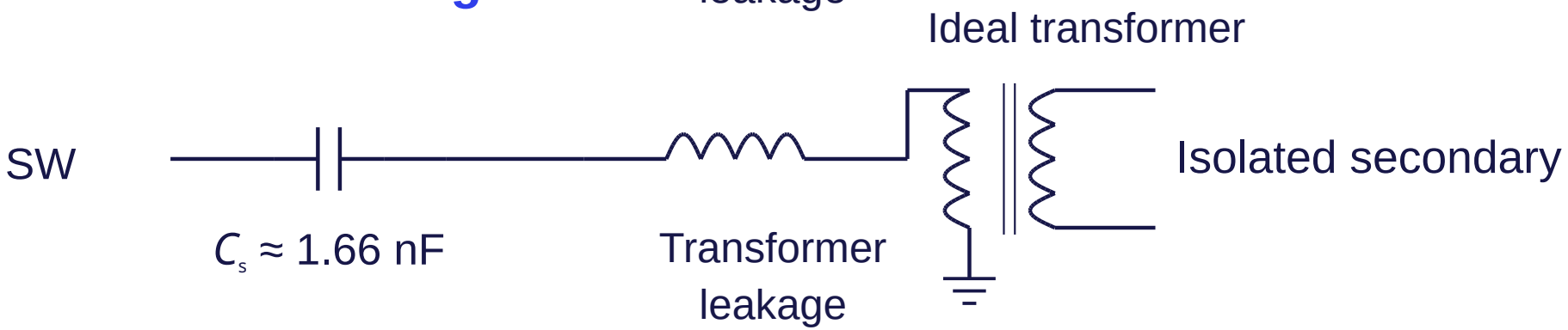
V2 removes the hot series inductor

Results

V2 before tank change



V2 after tank change



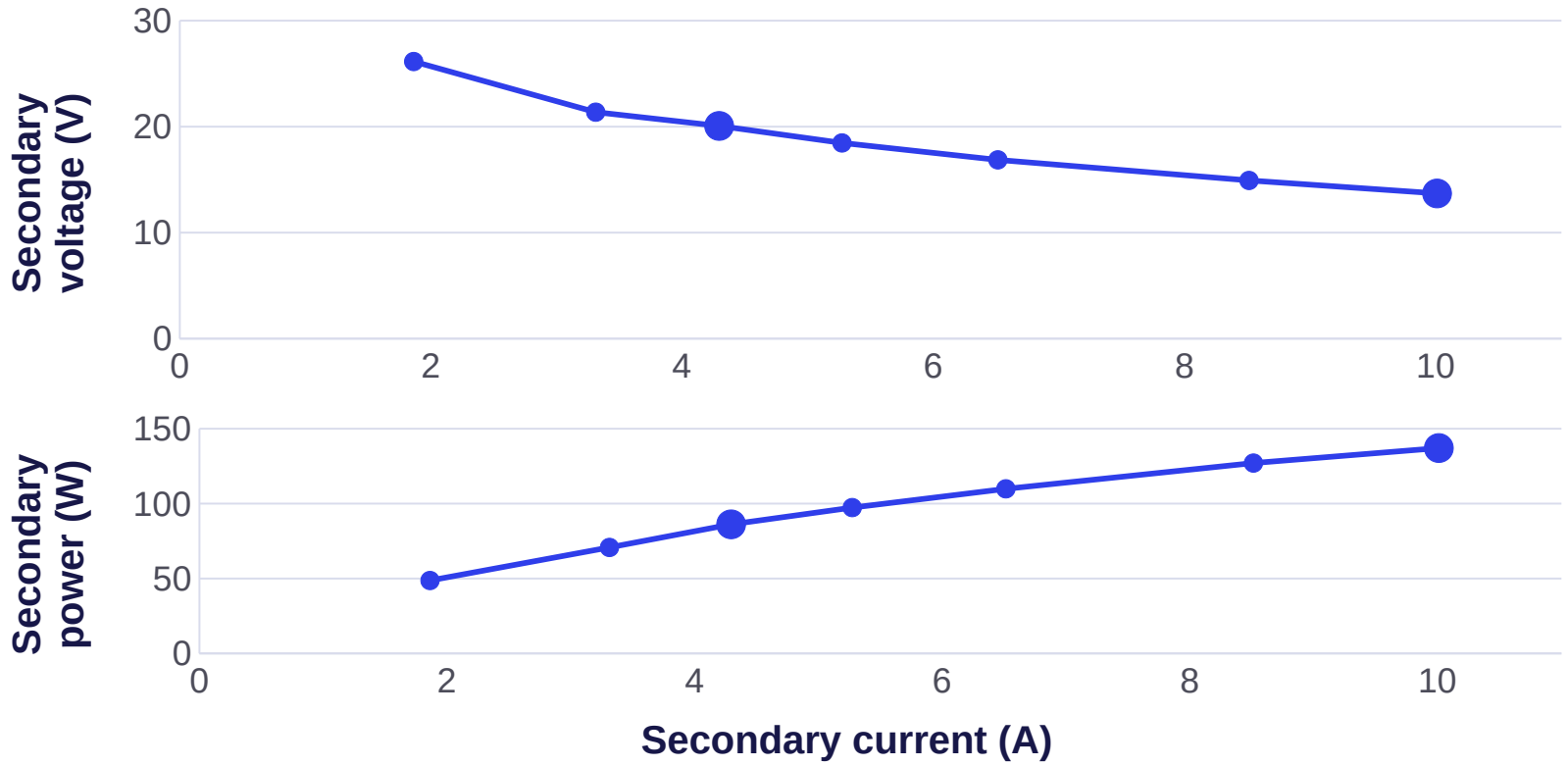
Software dead time ≈ 60 ns

Half-bridge heating fell; input bridge remained hot

Higher secondary power came with voltage droop

Results

V2 · direct secondary load before USB-C PD · 240 V AC, 50 Hz



20.07 V at 4.29 A

13.69 V at 10.01 A

86.17 W at 4.29 A

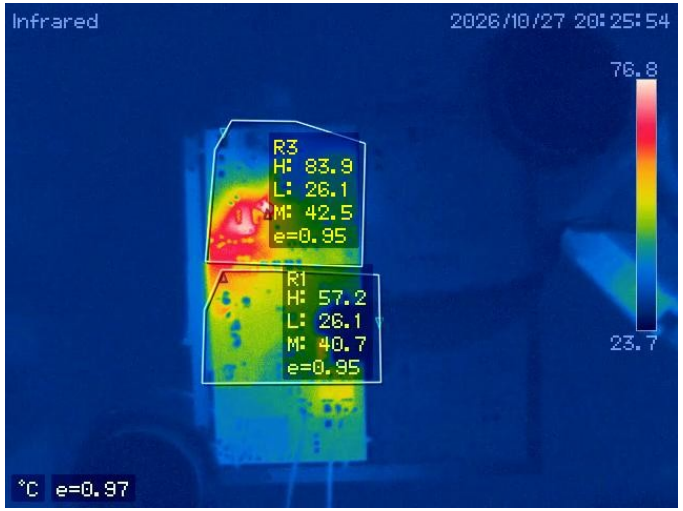
137.08 W at 10.01 A

Secondary and bridge heating guide the redesign

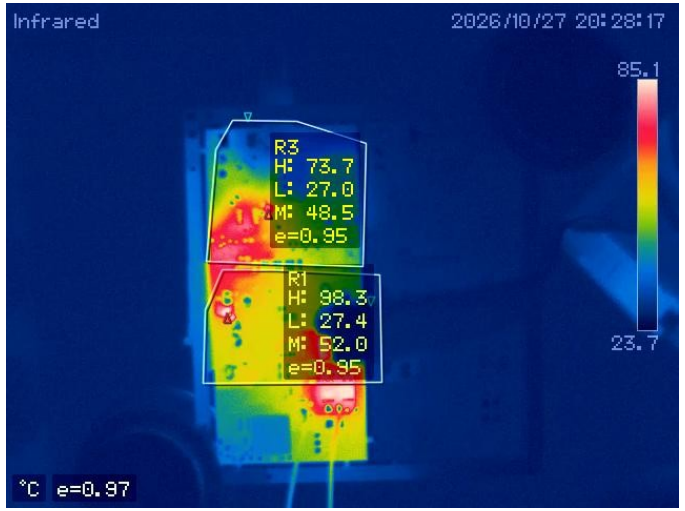
Results

V2 · Electronic load at secondary output

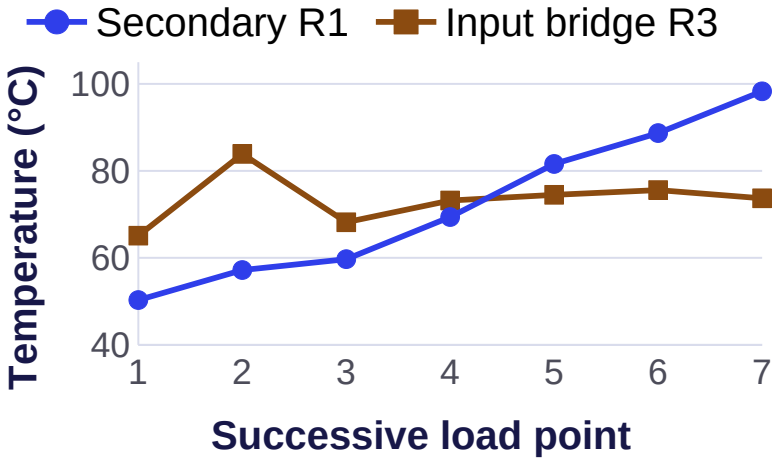
70.75 W at secondary



137.08 W at secondary



Maximum surface temperature by region



Input bridge R3: 83.9 °C

Secondary R1: 98.3 °C

No load point reached thermal equilibrium

Design direction: reduce winding and rectifier loss; widen bridge current and heat paths

Controlled USB-C output, with voltage and thermal limits

Conclusion

Implementation	USB-C integration and secondary-voltage burst control
Connector result	V1: 79.89 W measured at the USB-C connector
Continuous power limit	100 W continuous remains limited by droop and heating

The tank was retuned around transformer leakage

Backup

Parameter	Original calculation	Modified V2 calculation
External series inductance	9.4 µH	Removed
Transformer leakage, mean	16.46 µH	16.46 µH
Total series inductance, L_{total}	25.86 µH	16.46 µH
Series capacitance, C_s	1.00 nF	1.66 nF
Ideal series resonance, f_0	0.990 MHz	0.963 MHz

$$f_0 = \frac{1}{2\pi\sqrt{L_{\text{total}} C_s}}$$

Ideal estimates; installed response also depends on load, magnetizing branch and parasitics.

V2 secondary-load measurements

Backup

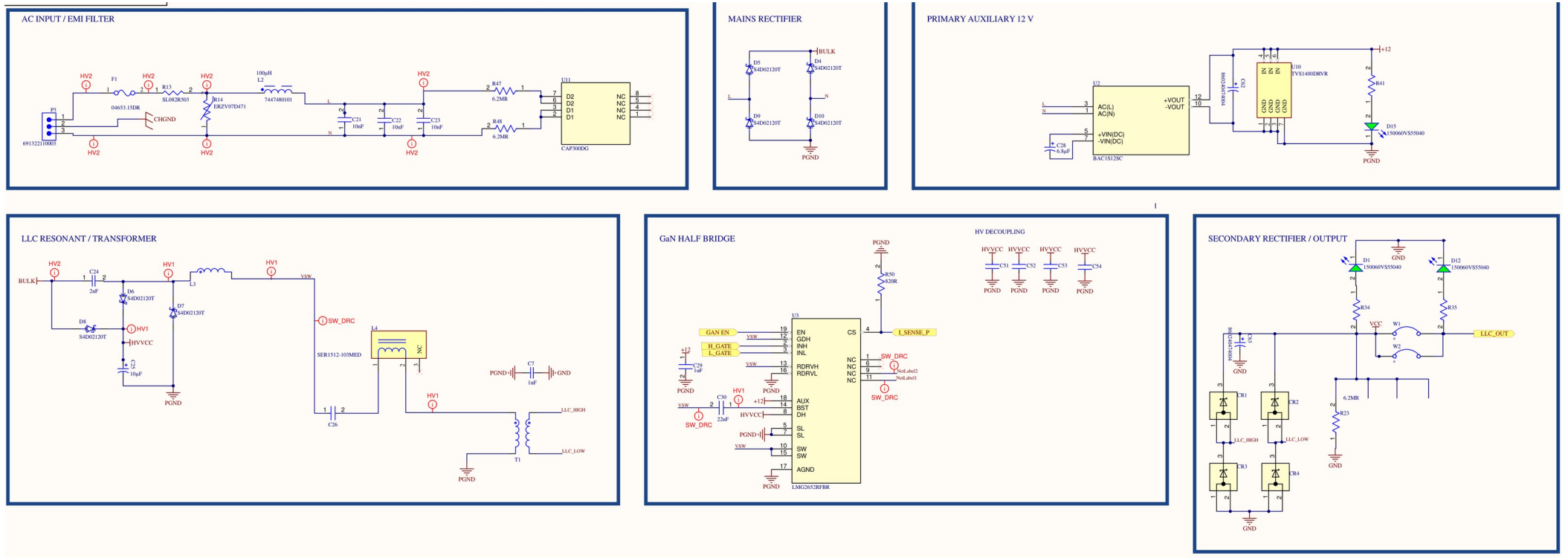
V2 · direct secondary load · 240 V AC, 50 Hz

Point	Secondary voltage (V)	Secondary current (A)	Secondary power (W)	R1 secondary max (°C)	R3 input bridge max (°C)
1	26.14	1.86	48.68	50.3	65.1
2	21.36	3.31	70.75	57.2	83.9
3	20.07	4.29	86.17	59.7	68.2
4	18.46	5.27	97.35	69.4	73.2
5	16.87	6.51	109.87	81.6	74.5
6	14.92	8.51	127.05	88.7	75.6
7	13.69	10.01	137.08	98.3	73.7

Successive load points; no thermal equilibrium

V2 power circuit and earlier work

Backup



V2 schematic before the tank change

Ammar et al., IEEE JESTPE, 2020 · doi: 10.1109/JESTPE.2020.2966143

Ammar et al., IEEE JESTPE, 2021 · doi: 10.1109/JESTPE.2021.3083404

V1 USB-C connector operating points

Backup

V1 · USB-C connector · approximately 240 V AC, 50 Hz

Selected (V)	Load (A)	Average VBUS (V)	Voltage error (%)	Connector power (W)
5	3.0	5.02118	+0.4236	15.06
9	3.0	9.00057	+0.0063	27.00
15	3.0	14.98350	−0.1100	44.95
20	3.7	19.99090	−0.0455	73.97
20	4.0	19.97320	−0.1340	79.89

19.9732 V × 4 A = 79.89 W at the connector